E-Commerce Project Proposal

PawMart

A One-Stop Web Marketplace for Pet Products & Pet Care Services

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Market Gaps & Proposed Solution

Fragmented Ecosystem

Pet owners currently navigate separate, disconnected platforms for retail shopping, health tracking, and scheduling vet care services.

Friction & Inefficiency

Complicated processes for scheduling routine services and high dependency on error-prone manual vaccination reminders.

Our Solution

A responsive web application integrating an e-commerce marketplace, service booking utility, and general husbandry assistant within a single cohesive system.



Project Objectives & Outcomes

Core Project Objectives

- **Consolidate Care:** Bridge fragmented pet retail and professional service domains under an integrated responsive portal.
- **Automate Logistics:** Establish lightweight, background alert systems to automate pet health/vaccine schedule management.
- **Innovate Feasibly:** Implement robust, rule-based assistive logic to optimize client purchasing and husbandry decisions.

Expected Deliverables

- **Operational MVP:** Delivery of a fully functional, multi-role web platform optimized for low-latency client and vendor use.
- **Stable Infrastructure:** Proof of a reliable deployment utilizing cost-free infrastructure suites (Vercel, Render, Atlas free tier).
- **Realistic Academic Scope:** A demonstrably sound software execution completed within a rigid 8-week developmental window.

Core Platform Features



Secure Authentication

Robust stateful role management using robust JSON Web Token (JWT) authorization alongside bcrypt password hashing protocols.



E-Commerce Core

Dynamic product inventory, intuitive shopping cart system, simulated checkouts, order processing, and a built-in user feedback system.



Dashboards & Control

Distinct interface architectures configured for Admins (marketplace moderation) and Vendors (product lists, inventory oversight).

Innovative Functional Modules



Smart Pet Care Assistant

Rule-based guidance utility offering structured feeding strategies, breed-specific care, and marketplace retail products based on target pet indicators.

**Provides general husbandry guidance only; non-veterinary diagnosis.*



Health & Vaccine Reminders

Digitized pet records archiving allergies, physical metrics, and age. Integrates background triggers executing automated programmatic email reminders.

**Automated via Nodemailer based on database timestamps.*



Appointment Booking

A lightweight scheduling system enabling clients to book appointments directly with clinical partners and grooming service vendors.

**Deterministic schedule allocation with individual owner dashboards.*

Strategic Technology Choices

Frontend

React.js + Tailwind CSS

Enables responsive design and rapid component rendering while maintaining zero UI framework style licensing fees.

Backend Server

Node.js + Express.js

Asynchronous, lightweight request handling that optimizes throughput for marketplace APIs and REST services.

Database Storage

MongoDB Atlas

Schemaless dynamic modeling perfect for heterogeneous product catalogs and complex pet health profile entities.

Media / Alerting

Cloudinary + Nodemailer

Simplifies rich image hosting optimization alongside lightweight transactional email delivery services at zero cost.

Host Environments

Vercel + Render

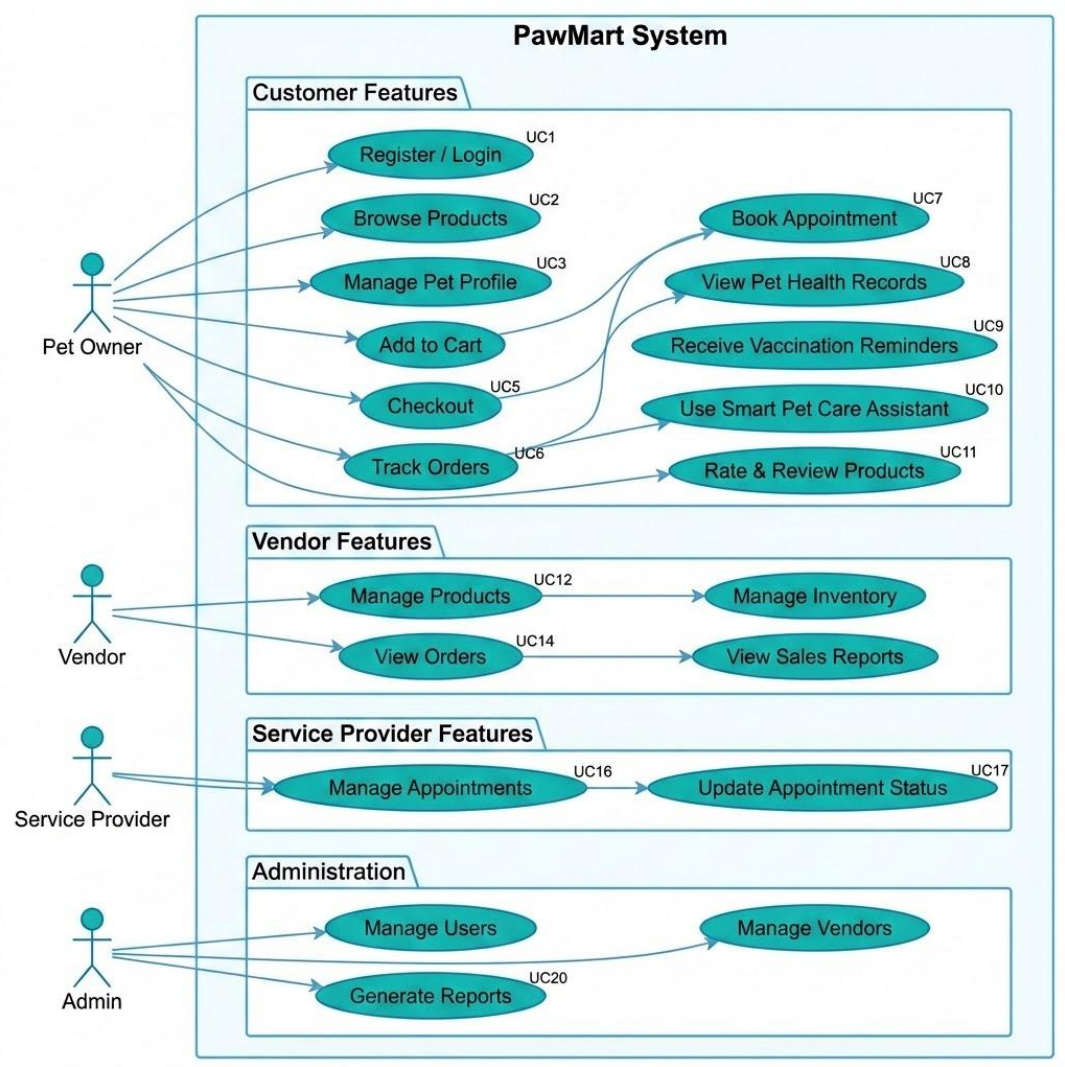
Seamless continuous deployment (CD) workflows optimized for client hosting and microserver runtimes.

Transaction Simulation

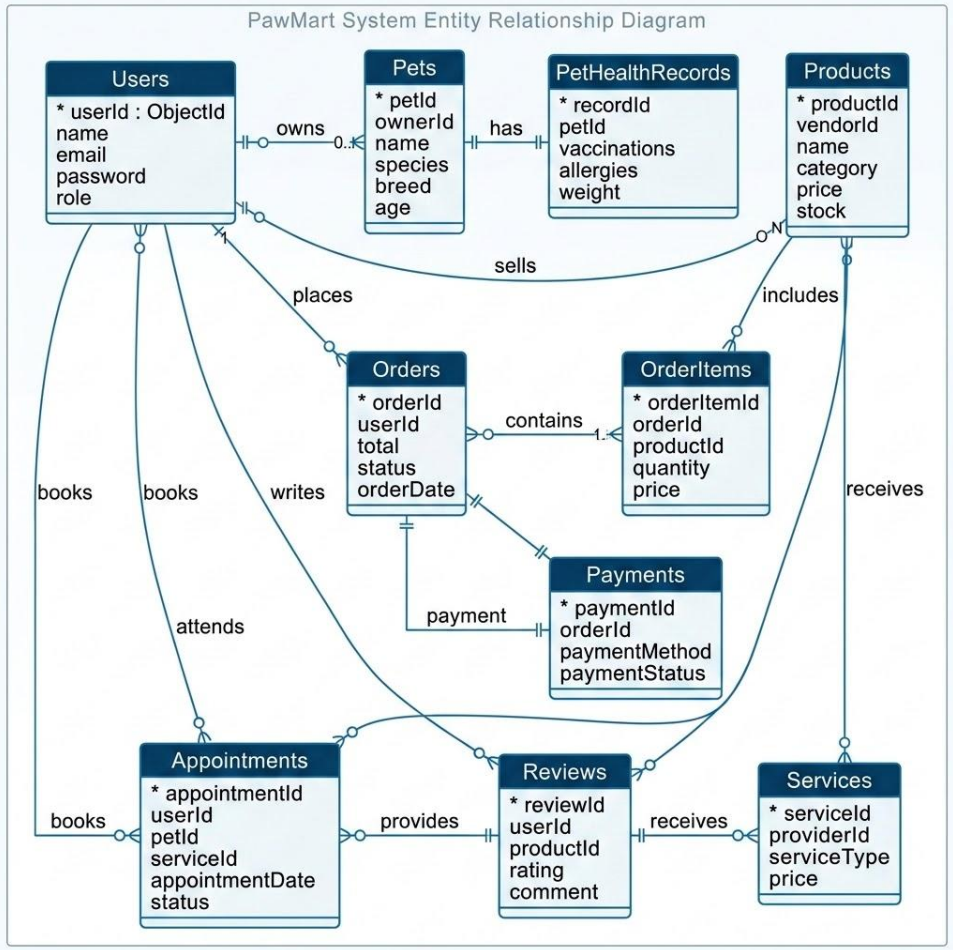
Simulated Payment Module

Eliminates complex third-party setup requirements (Stripe, Twilio) and compliance constraints for our academic MVP.

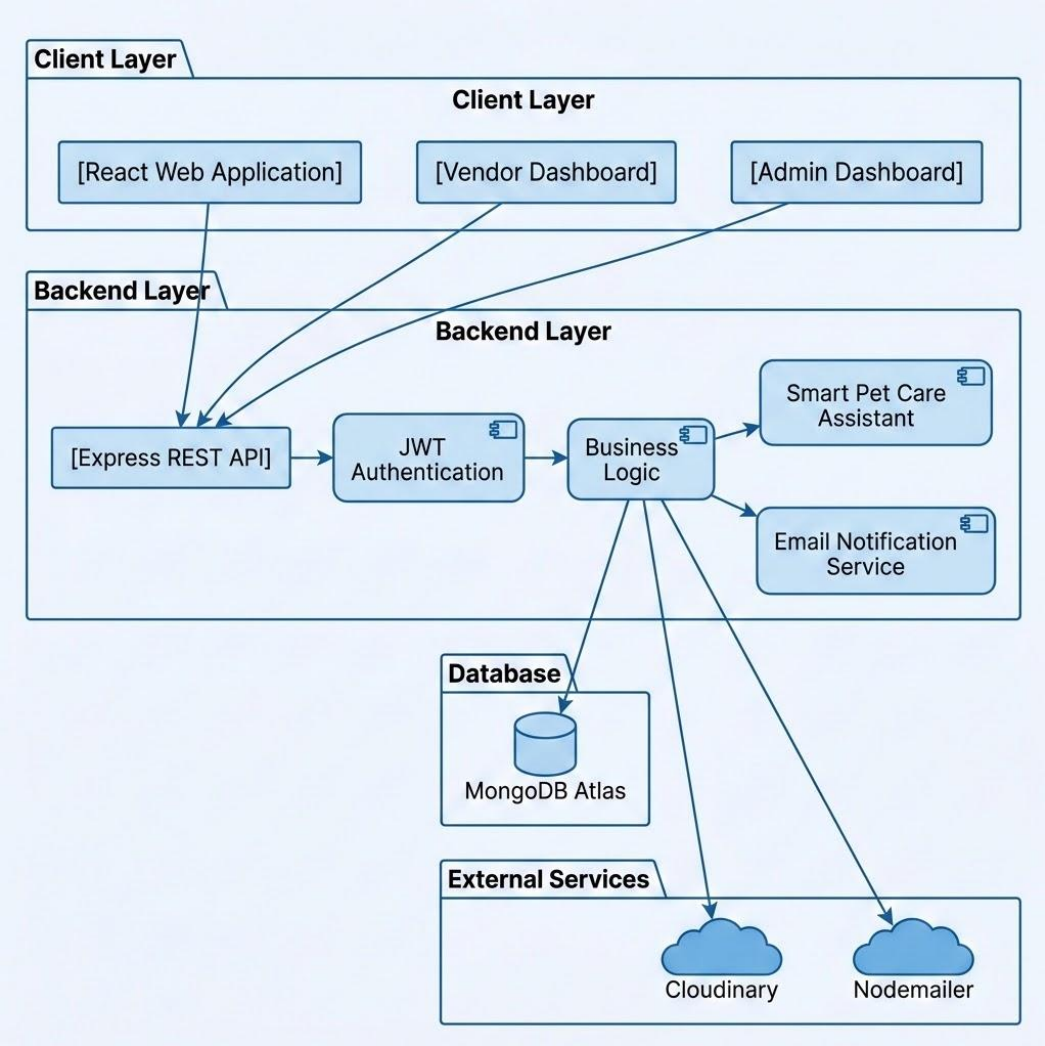
Use Case Diagram



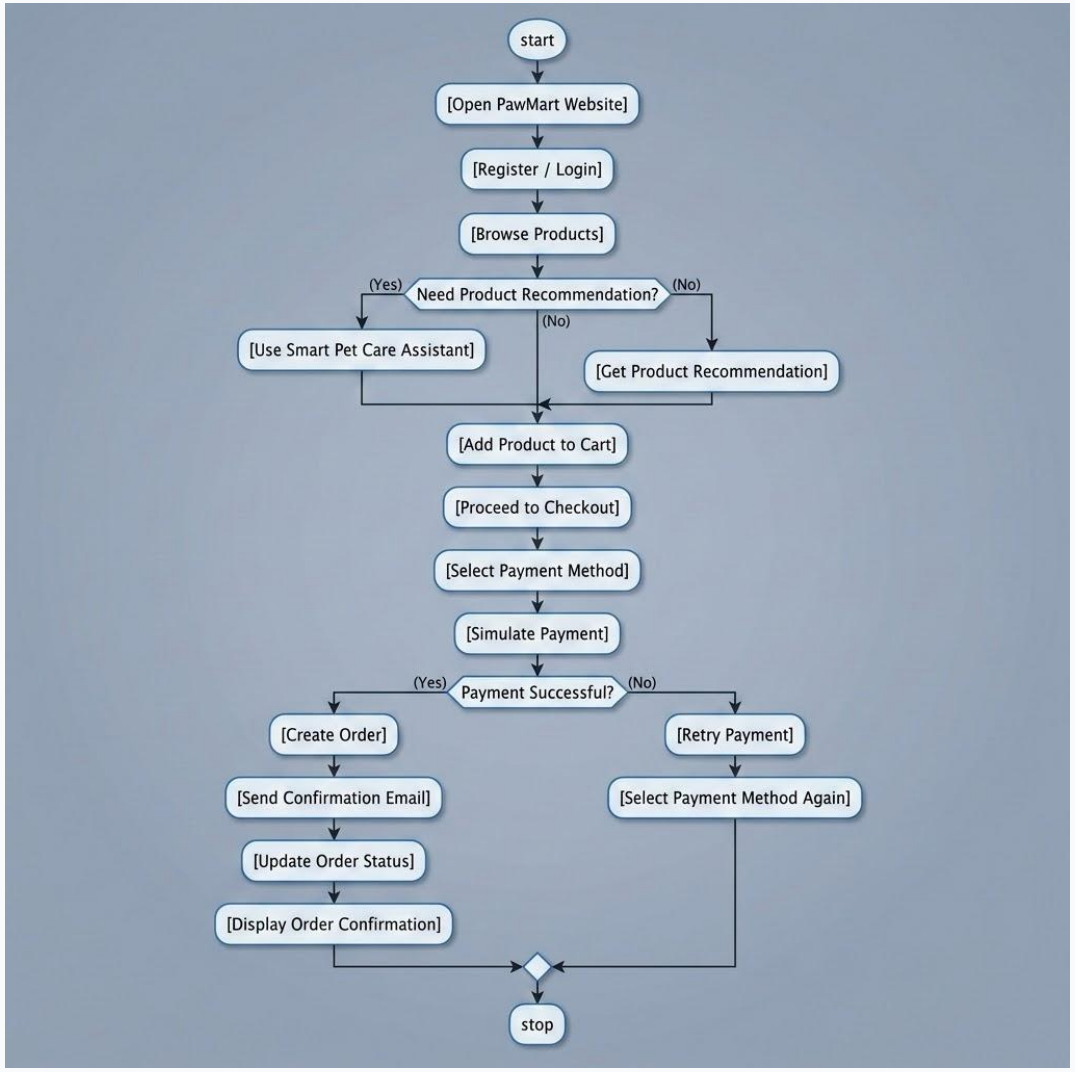
Entity Relationship Diagram (ERD)



System Architecture Diagram



Activity Diagram



Database Design & Justification

DATABASE ENGINE

MongoDB Atlas

Free Tier NoSQL

CORE COLLECTIONS

Users, Pets, PetHealthRecords, Products, Orders, OrderItems, Payments, Services, Appointments, Reviews

Why MongoDB was selected

Document Flexibility

Stores heterogeneous catalog attributes (such as medical alerts, varied sizing/weights, and review objects) natively in convenient BSON formats.

MERN-Native Synergy

Seamless document-to-JSON integration with the Node/Express backend, reducing database query transformation overhead.

Scalable Cloud Setup

Easily provisions a free tier cluster natively on AWS with zero development server configuration or licensing costs.

Team Responsibilities & Ownership

Member Name	Core Modules & Responsibilities
V. Prashoharan ITBIN 2211 0264	Authentication, Authorization, Admin Dashboard
Bashini Rashmika Vindyani ITBIN 2211 0122	Product Management, Vendor Dashboard
J.P. Inuka Sithanja Jayasinghe ITBIN 2211 0324	Shopping Cart, Checkout, Simulated Payment Module
D.S.N. Vitharana ITBIN 2211 0312	Order Management, Appointment Booking
B.R.G.S Sandaruwan ITBIN 2211 0278	Reviews & Ratings, Smart Pet Care Assistant, Pet Health Records & Vaccination Reminders

8-Week Development Timeline

Week 1
Requirements analysis, UI mockups, database collection planning.

Week 2
Authentication setup, JWT middleware, Atlas DB initialization.

Week 3
Product management models, Vendor dashboards.

Week 4
Shopping cart mechanisms, Checkout logic workflow.

Week 5
Mock Payment Engine, order verification schemas.

Week 6
Appointment booker, Pet health profile dashboards.

Week 7
Smart Care Assistant, rating modules, automated alerts & testing.

Week 8
Integration, bug verification, Render/Vercel deployment, slides preparation.

Thank You!

Questions & Discussion

PawMart Project Team

